Problem Statement

Imagine that your group is a professional software engineering team. You have been hired to meet the needs of a client who would like to make their business more sustainable (https://sdgs.un.org/goals).

Businesses can leverage software in various ways to become more sustainable. Here are some examples, courtesy of ChatGPT:

1. **Energy Management Software**: Implementing energy management software helps businesses monitor and optimise their energy consumption. This software can track energy usage in real time, identify areas of inefficiency, and suggest strategies for reducing energy consumption, ultimately leading to cost savings and environmental benefits.
2. **Supply Chain Management Software**: Sustainable supply chain management software allows businesses to track and manage their supply chains more efficiently. It can help identify suppliers with environmentally friendly practices, track the carbon footprint of products throughout the supply chain, and optimise transportation routes to reduce emissions.
3. **Green Building Software**: For businesses with physical facilities, green building software can be used to design and manage environmentally friendly buildings. This software helps optimise building designs for energy efficiency, track resource consumption, and manage waste more effectively.
4. **Virtual Collaboration Tools**: Virtual collaboration tools enable remote work and reduce the need for employees to commute to a physical office, thereby lowering carbon emissions associated with transportation. These tools include video conferencing software, project management platforms, and document-sharing platforms.
5. **Waste Management Software**: Waste management software helps businesses track and manage their waste generation more effectively. It can facilitate waste tracking, recycling initiatives, and compliance with waste management regulations, leading to reduced waste generation and lower environmental impact.
6. **Sustainable Product Design Software**: Software tools for sustainable product design assist businesses in creating environmentally friendly products. These tools analyse product lifecycle impacts, assess materials for sustainability, and optimise product designs for reduced environmental footprint.
7. **Data Analytics for Sustainability Reporting**: Businesses can use data analytics software to analyse and report on their sustainability performance.

These tools help collect and analyse sustainability data, track progress towards sustainability goals, and generate reports for stakeholders and regulatory compliance.

8. **Renewable Energy Management Software**: For businesses investing in renewable energy sources such as solar or wind power, renewable energy management software can help optimise energy production and consumption. These tools monitor renewable energy systems, forecast energy generation, and optimise energy usage to maximise the benefits of renewable energy investments.

For more ideas, you could read this article: What are the most pressing world problems?

https://80000hours.org/problem-profiles/?utm_source=Live+Audience&utm_campaign=d9ad4e8e7d-briefing-dy-20240226&utm_medium=email&utm_term=0_b27a691814-d9ad4e8e7d-48942559

By integrating these software solutions into their operations, businesses can streamline processes, reduce resource consumption, minimise waste, and lower their environmental impact, ultimately contributing to a more sustainable future.

The business owners are very new to the idea of becoming more sustainable. They have hired your team so that you can make recommendations to them. You should be innovative and creative in your approach to this project.

Pick one of the UN 17 Sustainable Development Goals (SDGs), and explain clearly how this project addresses the Goal.

Special notes:

The focus of COMP3030J is on developing your project management and soft skills. It is strongly recommended that you use the programming and web development skills you already know instead of learning a new framework for your project. In addition to your programming and web development skills, you have completed modules in Distributed Systems, Mobile Computing, Computer Graphics, Computer Networks, Databases and Information Systems. We would love to see you demonstrate these skills in your projects.

During the assessment of your projects, we will take into consideration the following:

- How well have you addressed the needs of the customer in terms of making their business more sustainable?
- Is your solution creative and innovative?

- Have you demonstrated the skills that you have learned in your other modules in your project?
- Is your project available for testing remotely? Is this straightforward?
- Have you demonstrated good project management throughout the semester?
- Have you demonstrated good teamwork throughout the semester?

Usernames/passwords for customer and staff accounts need to be supplied for testing purposes. Although you may have a way for new customers to register, etc., this should not be the only way that your solution is tested. In other words, when we test your solution, we should not need to register new accounts.

Please keep in mind that whatever software you develop needs to be available for testing remotely i.e. from Ireland. You must also use the VMs that we are providing you with to host any software that you develop. We will discuss your ideas with you as the semester progresses.

Use of Generative AI

Below, you will find the AI Assessment Scale used in the UCD College of Science. Different levels of AI use are available. In our module, we will use Level 3: "AI Collaboration". This means that you will be asked to discuss how you have used Generative AI and to demonstrate how you have critically evaluated and modified any AI-generated content.

The AI Assessment Scale

1	**NO AI**	The assessment is completed entirely without AI assistance, ensuring that students rely solely on their existing knowledge, understanding, and skills. **You must not actively use AI at any point during the assessment. You must demonstrate your core skills and knowledge.**
2	**AI PLANNING**	AI may be used for initial research (i.e. activities such as brainstorming, outlining, and information gathering collation). This level focuses on the effective use of AI for planning, synthesis, and ideation, but students complete the assessment independently by developing, articulating and refining these ideas. **You may use AI for planning, idea development, and research. Your engagement with AI stops when you begin to create your submission. Your final submission should therefore be authored entirely by yourself and should show how you have developed and refined these ideas.** **You should keep a comprehensive record of all outputs generated by AI, and may be required to document these as part of the activities, or present them on demand.**
2X	**AI REVIEW**	AI may be used in the same way as for Level 2, with the addition that an appropriate AI tool may be used to correct grammatical errors and improve the language of text you alone have authored. You must not allow the AI tool to alter the sense of any text, offer a critique of or modify any arguments. The assessor may indicate the specific instruction you are to provide the AI tool with.
3	**AI COLLABORATION**	AI may be used during initial research plus drafting, feedback, and refinement.. Students should critically evaluate and modify the AI suggested outputs, demonstrating their understanding. **You may use AI to comment on and review text you have authored. AI is intended to act akin to an instructor's comments on an essay or thesis chapter. assist with specific tasks such as drafting text, refining and evaluating your work. You are strongly advised to critically evaluate the outputs AI provides. must critically evaluate and modify any AI-generated content you use.** **You should keep a comprehensive record of all outputs generated by AI, and may be required to document these as part of the activities, or present them on demand.**
4	**FULL AI**	AI may be used to complete any elements of the task, with students directing AI to achieve the assessment goals. Assessments at this level may also require engagement with AI to achieve goals and solve problems. **You may use AI extensively throughout your work either as you wish, or as specifically directed in your assessment. Focus on directing AI to achieve your goals while demonstrating your critical thinking.** **You should keep a comprehensive record of all outputs generated by AI, and may be required to document these as part of the activities, or present them on demand.**
5	**AI EXPLORATION**	AI is used creatively to enhance problem-solving, generate novel insights, or develop innovative solutions to solve problems. Students and educators co-design assessments to explore unique AI applications within the field of study. **You should use AI creatively to solve the task, potentially co-designing new approaches with your instructor.**